16 Bit CPU Project

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Instruction Set Architecture (ISA)

8 Registers, 2-Operand Format, Word Addressable, Single Cycle

- **15-12**: Condition bits
    - Identical to ARM condition set
- **11-10**: Operation code bits
    - **00** - Data processing
    - **01** - Memory
    - **10** - Branch
    - **11** - Special
- **9-0**: Class specific bits

Data Processing Bits

- **9-6**: Function
- **5-3**: Destination Register/First Operand (R_d)
- **2-0**: Second Operand (R_m)

Memory Bits

- **9**: Load/Store Bit
    - **0**: Load (LDR)
    - **1**: Store (STR)
- **8-6**: Destination Register (R_d)
- **5-3**: Source Register/Address Register (Ra_m)
- **2-0**: Offset Register (R_{off})
    - Interpreted as two's complement (signed)
    - Added to source register

Branch Bits

- **9**: B/BL Bit

  - **0**: Branch (B)
  - **1**: Branch and Link (BL)
- **8-0**: Offset
  - Interpreted as two’s complement (signed)
  - PC-relative

Special Bits

- **9-8**: Function
- **7-6**: 11
- **5-3**: Destination/Source Register (R_d)
- **2-0**: 111